Exact conditional min-entropy of two-qubit unitary channels with white noise

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Abstract

The conditional min-entropy of a bipartite channel involves a trace-preserving marginal constraint absent from the corresponding state optimization. For an arbitrary two-qubit unitary mixed with the maximally mixed replacer, we prove that this constraint does not change the optimum. If ρ_U is the conditioning-side reduction of the normalized pure unitary Choi state and $L = \operatorname{Tr}\sqrt{\rho_U}$, both semidefinite objectives equal $(1-p)L^2 + p/4$, where p is the replacer weight. A common optimizer is $(1-p)L\sqrt{\rho_U} + (p/16)\mathbb{I}_4$. Matching positive witnesses establish the expression even when ρ_U is singular; the known square-root marginal property of locally maximally mixed two-qubit states makes the optimizer channel-feasible. This resolves the two-qubit white-noise extension posed by Badhani and Das (arXiv:2604.01217v2), using their noiseless result and qubit marginal lemma. The conclusion concerns single-use min-entropy for this specified noise family.

1 Introduction

Conditional channel entropies quantify quantum processes in the presence of a conditioning process. Their optimization differs from the state problem because the comparison operator must correspond to a trace-preserving channel [1, 2]. For conditional min-entropy this distinction is a linear marginal constraint in a semidefinite program (SDP). Understanding when that constraint costs nothing allows a channel quantity to be evaluated directly from a normalized Choi state.

Badhani and Das prove equality for noiseless two-qubit unitaries in Proposition 10 of Ref. [2]. Their Figure 2 reports the same behavior for identity, CNOT and SWAP with white noise, and the following paragraph asks whether it holds for arbitrary noisy two-qubit unitaries. We establish that two-qubit extension and give the exact objective and a common optimizer for every noise weight. The essential ingredients are an explicit primal–dual certificate for a pure projector plus scalar noise, and the qubit square-root marginal lemma already given in Appendix D of Ref. [2]. The pure-state min-entropy formula [3], qubit normal forms and SDP averaging are established tools; no new general symmetry principle is needed.

The statement of the open extension persists in version 2 of the source. A bounded primary-literature comparison through 12 October 2026 found no later resolution of this exact channel identity. This comparison does not establish exhaustive priority for the intermediate state formula. The tele-covariant von Neumann entropy identity in Ref. [1] is a different result, and its examples mixing a gate with the identity channel use a different noise map.

2 Model and normalization

All four ports A', B', A, B are qubits. For $U \in U(4)$ and $0 \le p \le 1$, define the completely positive, trace-preserving map

$$\mathcal{N}_{p,U}(Z) = (1-p)UZU^\dagger + p\,\mathrm{Tr}(Z)\frac{\mathbb{I}_{AB}}{4}. \tag{1}$$

Thus p denotes the white-noise weight, not the unitary weight. Let Φ_2 be the projector on $(|00\rangle + |11\rangle)/\sqrt{2}$, and set

$$\Omega = (\mathrm{id}_{R_A R_B} \otimes \mathcal{N}_{p,U})(\Phi_{R_A A'} \otimes \Phi_{R_B B'}). \tag{2}$$

We group factors as $X = R_A A$ and $Y = R_B B$, each of dimension four. With this ordering,

$$\Omega = (1-p)|u\rangle\langle u| + \frac{p}{16}\mathbb{I}_{XY}, \qquad u_{r_a,a,r_b,b} = \frac{1}{2}U_{2a+b,\,2r_a+r_b}. \tag{3}$$

Here $r_a, a, r_b, b \in \{0,1\}$ and $\langle u|u\rangle = 1$. In particular $\mathrm{Tr}\,\Omega = 1$. Define the reduction of the *pure unitary* Choi state by

$$\rho_U = \mathrm{Tr}_X\,|u\rangle\langle u|, \qquad L = \mathrm{Tr}\,\sqrt{\rho_U}. \tag{4}$$

The distinction between ρ_U and $\mathrm{Tr}_X\,\Omega$ is essential.

The two objectives under comparison are

$$s(\Omega) = \min_{Q \succeq 0}\{\mathrm{Tr}\,Q : \mathbb{I}_X \otimes Q \succeq \Omega\}, \tag{5}$$

$$t(\Omega) = \min_{Q \succeq 0}\left\{\mathrm{Tr}\,Q : \mathbb{I}_X \otimes Q \succeq \Omega,\ \mathrm{Tr}_B\,Q = \frac{\mathrm{Tr}\,Q}{2}\mathbb{I}_{R_B}\right\}. \tag{6}$$

The state conditional min-entropy is $H_{\min}(X|Y)_\Omega = -\log_2 s(\Omega)$ [3]. In the channel convention of Refs. [1, 2],

$$S_\infty[A|B]_{\mathcal{N}} = -\log_2 t(\Omega) - 1. \tag{7}$$

For clarity, Eq. (87) of Ref. [1] uses the unnormalized Choi operator $\Gamma_{\mathcal{N}} = 4\Omega$. Its variable $M = 4Q$ has objective $\mathrm{Tr}\,M/|B'| = 2\,\mathrm{Tr}\,Q$. Its marginal constraint becomes exactly that in Eq. (6), which explains the one-bit offset in Eq. (7). Scalar identity operators give feasible points, and trace-bounded positive operators form a compact set, so the minima are attained.

3 Exact objective and common optimizer

Theorem 1. *For every $U \in U(4)$ and every $p \in [0,1]$, the objectives in Eqs. (5)–(6) satisfy*

$$s(\Omega) = t(\Omega) = (1-p)L^2 + \frac{p}{4}. \tag{8}$$

A common optimizer is

$$Q_* = (1-p)L\sqrt{\rho_U} + \frac{p}{16}\mathbb{I}_Y. \tag{9}$$

Consequently,

$$S_\infty[A|B]_{\mathcal{N}_{p,U}} = H_{\min}(X|Y)_\Omega - 1 = -\log_2\left((1-p)L^2 + \frac{p}{4}\right) - 1. \tag{10}$$

The result includes zero Schmidt coefficients, repeated eigenvalues, and both noise endpoints.

Proof. First obtain matching bounds for the state problem. Write the positive-coefficient Schmidt decomposition as

$$|u\rangle = \sum_{i=1}^{r} c_i |x_i\rangle|y_i\rangle, \qquad c_i > 0, \quad \sum_i c_i^2 = 1.$$

Then $L = \sum_i c_i$ and $\sqrt{\rho_U} = \sum_i c_i |y_i\rangle\langle y_i|$. For an arbitrary vector z, weighted Cauchy–Schwarz gives

$$|\langle u|z\rangle|^2 \le L \sum_i c_i |\langle x_i y_i|z\rangle|^2 \tag{11}$$

$$\le \langle z|(\mathbb{I}_X \otimes L\sqrt{\rho_U})|z\rangle. \tag{12}$$

The second inequality includes all components along X for each y_i. Therefore $\mathbb{I}_X \otimes L\sqrt{\rho_U} \succeq |u\rangle\langle u|$. Put $\alpha = 1 - p$ and $\beta = p/16$. Equation (9) is state-feasible because its slack is $\alpha(\mathbb{I}_X \otimes L\sqrt{\rho_U} - |u\rangle\langle u|)$, and its trace is $\alpha L^2 + 4\beta$.

For a lower bound let

$$\Pi = \sum_{i=1}^{r} |y_i\rangle\langle y_i|, \quad |w\rangle = \sum_{i=1}^{r} |x_i\rangle|y_i\rangle, \quad W = |w\rangle\langle w| + \frac{\mathbb{I}_X}{4} \otimes (\mathbb{I}_Y - \Pi). \tag{13}$$

We have $W \succeq 0$, $\mathrm{Tr}_X W = \mathbb{I}_Y$, $\mathrm{Tr}\, W = 4$, and $\langle u|W|u\rangle = L^2$. For every state-feasible Q,

$$\mathrm{Tr}\, Q = \mathrm{Tr}[(\mathbb{I}_X \otimes Q)W] \ge \mathrm{Tr}(\Omega W) = \alpha L^2 + 4\beta. \tag{14}$$

This proves $s(\Omega) = \alpha L^2 + 4\beta$ by weak duality. The support-complement term in Eq. (13) is needed for $r < 4$ and $p > 0$: omitting it loses $(4 - r)\beta$ from the lower bound. No inverse or full-rank assumption has been used.

It remains to show that Q_* obeys the channel marginal. Unitarity gives both full reference and full output marginals $\mathbb{I}_4/4$ for $|u\rangle\langle u|$. Tracing the other qubit then gives

$$\mathrm{Tr}_B\, \rho_U = \frac{\mathbb{I}_{R_B}}{2}, \qquad \mathrm{Tr}_{R_B}\, \rho_U = \frac{\mathbb{I}_B}{2}. \tag{15}$$

The square-root marginal lemma for such two-qubit states is Appendix D, Lemma 9 of Ref. [2]. For completeness, expand

$$\rho_U = \frac{1}{4}\left(\mathbb{I} \otimes \mathbb{I} + \sum_{i,j=1}^{3} T_{ij}\sigma_i \otimes \sigma_j\right), \qquad T_{ij} \in \mathbb{R}.$$

There are no single-Pauli terms by Eq. (15). A real singular-value decomposition, with reflection signs absorbed in the diagonal entries, diagonalizes T using two proper rotations. Local qubit unitaries implement these rotations. In that local basis ρ_U is a linear combination of $\mathbb{I} \otimes \mathbb{I}$, $\sigma_x \otimes \sigma_x$, $\sigma_y \otimes \sigma_y$ and $\sigma_z \otimes \sigma_z$. These commute and share the four Bell vectors as an eigenbasis. Its positive square root is therefore a sum of Bell projectors with weights $\sqrt{\lambda_j}$. Each projector has marginal $\mathbb{I}_2/2$. Returning to the original basis yields

$$\mathrm{Tr}_B \sqrt{\rho_U} = \frac{L}{2}\mathbb{I}_{R_B}. \tag{16}$$

Zero or repeated eigenvalues do not affect this spectral argument. Using $\mathrm{Tr}_B\, \mathbb{I}_Y = 2\mathbb{I}_{R_B}$,

$$\mathrm{Tr}_B\, Q_* = \left(\frac{\alpha L^2}{2} + 2\beta\right)\mathbb{I}_{R_B} = \frac{\mathrm{Tr}\, Q_*}{2}\mathbb{I}_{R_B}.$$

Thus Q_* is feasible for t. Feasible-set inclusion gives $s \le t \le \mathrm{Tr}\, Q_* = s$, proving the assertion. All inequalities remain valid for $p = 0$ and $p = 1$. □

3.1 Why the marginal constraint has zero cost

The same equality has a structural averaging interpretation. Choose a local-unitary Bell eigenbasis $\{y_i\}_{i=1}^4$ of ρ_U and corresponding Schmidt vectors $\{x_i\}$, completed orthonormally when an eigenvalue is zero. For each sign vector $\varepsilon \in \{-1, 1\}^4$, define

$$V_X(\varepsilon) = \sum_i \varepsilon_i |x_i\rangle\langle x_i|, \qquad V_Y(\varepsilon) = \sum_i \varepsilon_i |y_i\rangle\langle y_i|.$$

Their tensor product fixes $|u\rangle$, and therefore fixes Ω. Conjugation maps every state-feasible Q to another feasible operator. Averaging over the sixteen signs preserves its trace and produces

$$\overline{Q} = \sum_i \langle y_i|Q|y_i\rangle |y_i\rangle\langle y_i|.$$

Every y_i has marginal $\mathbb{I}_2/2$, so $\overline{Q}$ obeys the channel constraint. This identifies the reason that the constraint has zero objective penalty. The averaging operations act on grouped spaces; their use in the proof does not assert their physical availability as local port operations.

4 Special gates, interpretation and limits

Let $\kappa = L^2$. Normalization and Cauchy–Schwarz give $1 \le \kappa \le 4$. Thus the gate dependence enters only through the Schmidt spectrum of its pure normalized Choi vector. The quantity $2^{-H_{\min}(X|Y)_\Omega}$ is affine in the replacer weight, although the entropy itself is logarithmic. The identity, CNOT and SWAP have $\kappa = 1, 2, 4$, respectively. Their objectives and endpoints are shown in Table 1. These are exact instances of Eq. (8), recovering the three examples of Ref. [2].

Table 1: Exact common objective $s = t$ for three gates, with p the maximally mixed replacer weight. Values are rational, not SDP solver estimates. Channel entropy is $-\log_2(s) - 1$.

Gate	κ	$p = 0$	$p = 1/2$	$p = 1$
Identity	1	1	5/8	1/4
CNOT	2	2	9/8	1/4
SWAP	4	4	17/8	1/4

For example, the noiseless SWAP has state entropy -2 bits and channel entropy -3 bits. The completely mixed replacer has state entropy $+2$ bits and channel entropy $+1$ bit for every U. These endpoints also check the normalized Choi convention and the one-bit channel offset.

The result removes the need to solve either SDP once the four eigenvalues of ρ_U are known. In the conditional athermality framework of Ref. [2], it evaluates the single-use channel min-entropy entering the trivial-Hamiltonian resource description. No additional thermodynamic conversion theorem is derived here. The proof relies on scalar white noise and a conditioning pair of qubits. It makes no extension to arbitrary mixed-unitary noise, higher dimensions, nontrivial Hamiltonians, continuous-time dynamics or regularized communication capacities. Neither the numerical gate checks nor the entropy sign should be interpreted as an entanglement-breaking criterion.

5 Methods and reproducibility

The result is exact and finite-dimensional. Its error control is the pair of analytical inequalities in the proof, with no perturbative, finite-size or floating-point approximation. The accompanying standard-library Python scripts reconstruct the Choi input in two tensor conventions

and test special-gate certificates with rational arithmetic. A separate exact complex-rational check covers eight unitary gates and four noise weights, including rank-deficient and complex instances, with 32 primal, dual and marginal certificates and 128 sign-invariance checks. The finite checks diagnose normalization and implementation errors; they do not establish the universal quantifiers. Regeneration instructions and the tested environment accompany the code. The self-contained proof is sufficient for the analytic result.

Generative AI was used in proposing and developing the argument, in writing reconstruction and certificate-checking code, in literature searches and comparison, in automated scientific checking, and in drafting this manuscript. These checks do not constitute human verification or external peer review.

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Data availability. No experimental data were used. The reproduction and exact certificate scripts, fixed test inputs and regeneration instructions are included in the accompanying source package.

References

[1] Siddhartha Das, Kaumudibikash Goswami, and Vivek Pandey. Conditional entropy and information of quantum processes. arXiv:2410.01740v2, 2024. Version 2, 30 October 2024.

[2] Himanshu Badhani and Siddhartha Das. Conditional channel entropy sets fundamental limits on thermodynamic quantum information processing. arXiv:2604.01217v2, 2026. Version 2, 23 June 2026.

[3] Robert König, Renato Renner, and Christian Schaffner. The operational meaning of min- and max-entropy. *IEEE Transactions on Information Theory*, 55(9):4337–4347, 2009.